

Excel Notes for Data Analysis & Interview Preparation

Source Notes Enhanced From User Practice Notes + Added Professional Data Analyst Concepts

Important Questions

Difference Between Workbook and Worksheet

Workbook	Worksheet
A complete Excel file	A single sheet inside a workbook
Can contain multiple sheets	Used to store data in rows and columns
Example: Sales_Report.xlsx	Example: January_Sales sheet

Real-World Example

A company may have one workbook containing:

- Sales Sheet
 - Employees Sheet
 - Finance Sheet
 - Dashboard Sheet
-

Difference Between Formula and Function

Formula

A formula is a calculation created manually using mathematical operators.

Example:

```
=A1+B1  
=A1*B1
```

Function

A function is a predefined formula already built into Excel.

Example:

```
=SUM(A1:A10)  
=AVERAGE(B1:B10)
```

Interview Tip

- Formula = User-created calculation
 - Function = Built-in Excel calculation
-

Difference Between Relative and Absolute Reference

Relative Reference

Cell reference changes automatically when formula is dragged.

Example:

```
=A1+B1
```

When dragged downward:

```
=A2+B2
```

Absolute Reference

Cell reference remains fixed using \$.

Shortcut:

```
F4
```

Example:

```
=$A$1
```

Real-World Use Case

Suppose tax rate is stored in one cell:

B1 = 10%

You want every employee salary multiplied with same tax rate.

Formula:

=A2*\$B\$1

Important Note

- Relative reference = Dynamic
- Absolute reference = Fixed
- Mixed reference also exists:

\$A1
A\$1

Chapter 1 & 2 — Excel Basics & Formatting

Save File With Password

Steps

File → Save As → Tools → General Options

Two types of password: 1. Password to Open File 2. Password to Modify File

Protect Specific Worksheet

Right-click Sheet → Protect Sheet

Used when:

- Sharing reports
- Preventing accidental editing
- Restricting formulas

Auto Fill / Serial Numbers

Methods

- Drag Fill Handle
- Double-click Fill Handle
- Select first 2 sequence values then drag
- Hold Ctrl while dragging for increment

Example

1
2

Drag → 3,4,5...

Cell Formatting

Shortcut:

Ctrl + 1	Alt + H + O + I
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Custom Date Formats

dd-mm-yyyy
mmm-yyyy
ddd-mmm-yyyy

Important Analyst Insight

Formatting changes only appearance, not actual value.

Conditional Formatting (Advanced)

Used to highlight cells/rows based on conditions.

Highlight Complete Row Dynamically

Steps

1. Select entire dataset without headers
2. Conditional Formatting → New Rule
3. Use formula to determine...
4. Select first cell of target column
5. Remove `$` from row reference if needed
6. Apply condition

Example

```
=C2="Sales"
```

Dynamic Version

Write department name in another cell:

```
=C2=$H$1
```

Now changing H1 changes highlighted rows dynamically.

Real-World Usage

- Highlight delayed projects
- Detect low-performing employees
- Highlight duplicate entries
- Detect negative profits

Interview Tip

Conditional Formatting is heavily used in dashboards.

Chapter 3 — Data Cleaning

Data Cleaning is one of the MOST IMPORTANT tasks in Data Analysis.

Real-world datasets are often:

- messy
- incomplete
- duplicated

- inconsistent

A Data Analyst spends huge time cleaning data before analysis.

Remove Duplicates

Steps

1. Select data
2. Data Tab
3. Remove Duplicates

Important Point

You can remove duplicates based on:

- one column
- multiple columns

Real-World Example

Removing duplicate:

- customer IDs
- transaction records
- emails

Interview Question

Why are duplicates dangerous?

Answer: Because they create incorrect analysis and wrong totals.

Remove Blank Rows

Steps

1. Ctrl + A
2. Ctrl + G
3. Special → Blanks
4. Ctrl + -
5. Entire Row

Why Important?

Blank rows can:

- break Pivot Tables
- affect formulas
- create import problems

Remove Unnecessary Spaces

Extra spaces cause major problems in analysis.

Example:

```
Mobile  
Mobile
```

Both LOOK same but are different values.

Problems Caused

- COUNTIF mismatch
- VLOOKUP failure
- Wrong grouping
- Incorrect Pivot results

Detect Using

```
=LEN(A2)
```

Remove Using

```
=TRIM(A2)
```

Unbreakable Space (CHAR 160)

TRIM cannot remove non-breaking spaces.

Solution

```
=SUBSTITUTE(A2,CHAR(160),"")
```

Combined Professional Formula

```
=TRIM(SUBSTITUTE(A2,CHAR(160),""))
```

Interview Tip

Many datasets imported from websites contain CHAR(160).

SUBSTITUTE Function

Used for replacing text.

Syntax

```
=SUBSTITUTE(text,old_text,new_text)
```

Example

```
=SUBSTITUTE(A2,"World","World")
```

Case Fixing

Functions

```
=UPPER(A2)  
=LOWER(A2)  
=PROPER(A2)
```

Use Cases

- Standardizing names
- Cleaning customer data
- Preparing reports

Paste as Values (Very Important)

When formulas depend on old columns, deleting original data may break formulas.

Solution

Copy → Paste as Values

Shortcut

Alt + H + V + V

Analyst Insight

Paste Values is one of the most used shortcuts in Data Analytics.

Replace Negative Values with Zero

Formula

=IF(A2<0,0,A2)

Real-World Example

Used when:

- negative stock not allowed
 - sales cannot be negative
 - cleaning financial datasets
-

Delete Irrelevant Columns

Always remove unnecessary columns before analysis.

Benefits:

- cleaner dataset
- faster analysis

- better performance
 - reduced confusion
-

Flash Fill

Shortcut:

Ctrl + E

Excel detects patterns automatically.

Examples

- Splitting names
- Formatting phone numbers
- Capitalization
- Extracting codes

Important Limitation

Flash Fill is NOT dynamic. If original data changes later, result does not update.

Dynamic Splitting Using Formulas

LEFT + FIND

=LEFT(A2,FIND("-",A2)-1)

RIGHT + LEN + FIND

=RIGHT(A2,LEN(A2)-FIND("-",A2))

Real-World Example

Splitting:

LHR-2025

Into:

- City
 - Code
-

Additional Important Text Functions

MID

Extract text from middle.

```
=MID(A2,2,5)
```

LEFT

Extract from left.

```
=LEFT(A2,3)
```

RIGHT

Extract from right.

```
=RIGHT(A2,4)
```

CONCAT

Combine text.

```
=CONCAT(A2," ",B2)
```

TEXTJOIN

Combine multiple values with delimiter.

```
=TEXTJOIN(", ",TRUE,A2:A10)
```

Data Validation

Used to control user input.

Examples

- 11-digit phone numbers
- dropdown lists
- restricting negative values

Real-World Usage

Prevents wrong data entry.

Interview Tip

Data Validation improves data quality.

Chapter 4 — Logical Functions

IF Function

Syntax

```
=IF(condition,true,false)
```

Example

```
=IF(D2>=50000,"Completed","Incomplete")
```

Nested IF

Used for multiple conditions.

Example

```
=IF(A2>=90,"A",  
IF(A2>=80,"B",  
IF(A2>=70,"C","Fail")))
```

Limitation

Too many nested IFs become difficult.

Professional analysts often prefer:

- XLOOKUP
- VLOOKUP
- SWITCH
- IFS

IFS Function

Cleaner alternative to nested IF.

```
=IFS(A2>=90,"A",A2>=80,"B",A2>=70,"C")
```

AND Function

Both conditions must be TRUE.

```
=IF(AND(B2="Finance",C2>=5),"Yes","No")
```

OR Function

At least one condition must be TRUE.

```
=IF(OR(A2>90,B2="Excellent"),"Selected","Rejected")
```

IFERROR (Very Important)

Used to hide formula errors.

Example

```
=IFERROR(VLOOKUP(A2,D:E,2,0),"Not Found")
```

Interview Tip

Professional reports should avoid ugly errors like:

```
#N/A  
#DIV/0!
```

Chapter 5 — Sort, Filter & Dynamic Arrays

Sort

Used to arrange data.

Types

- A to Z
- Z to A
- Smallest to Largest
- Multi-level sorting

Real-World Example

Sort by:

- City
 - Revenue
 - Department
 - Date
-

Filter

Displays only required data.

Real-World Example

- Show only Lahore sales
 - Show only completed orders
 - Show only 2025 data
-

SORT Function

```
=SORT(A2:J1001,2,1,TRUE)
```

FILTER Function

```
=FILTER(A2:J1001,D2:D1001=AA2,"Not Found")
```

Important

Requires empty spill area. Otherwise shows:

```
#SPILL!
```

TAKE Function

```
=TAKE(A2:J1001,5)
```

Returns only selected rows/columns.

Combined Dynamic Formula

```
=TAKE(SORT(FILTER(A2:J1001,C2:C1001="Mumbai","Not Found"),2,1),5)
```

What It Does

1. Filters Mumbai rows
 2. Sorts them
 3. Returns top 5 rows
-

UNIQUE Function (Important)

Returns unique values.

```
=UNIQUE(A2:A100)
```

Real-World Use

- Unique cities
 - Unique products
 - Unique employees
-

Chapter 6 — Lookup Functions

VLOOKUP

Vertical lookup.

Searches left-most column and returns value from another column.

Syntax

```
=VLOOKUP(lookup_value,table_array,col_index_num,range_lookup)
```


Example

```
=VLOOKUP(D2,emp_table,2,TRUE)
```

Important VLOOKUP Rules

Exact Match

```
FALSE or 0
```

Approximate Match

```
TRUE or 1
```

Important

Approximate match requires ascending order.

VLOOKUP Limitations

- Searches only left to right
 - Breaks if columns inserted
 - Slower on huge datasets
 - Less flexible
-

MATCH Function

Returns position number.

Syntax

```
=MATCH(value,array,0)
```

INDEX + MATCH

Professional replacement for VLOOKUP.

Formula

```
=INDEX($A$1:$J$9,  
MATCH(B15,$A$1:$A$9,0),  
MATCH(C14,$A$1:$J$1,0))
```

Advantages Over VLOOKUP

- Faster
- Flexible
- Works left and right
- More reliable

Interview Question

Why use INDEX MATCH instead of VLOOKUP?

Answer: Because INDEX MATCH is more flexible and efficient.

XLOOKUP (VERY IMPORTANT)

Modern replacement for VLOOKUP.

Syntax

```
=XLOOKUP(lookup_value,lookup_array,return_array)
```

Example

```
=XLOOKUP(A2,D:D,E:E,"Not Found")
```

Advantages

- Searches both directions
- No column index needed
- Built-in error handling
- Easier syntax

Interview Tip

Learning XLOOKUP is strongly recommended.

Chapter 7 — Statistical Functions

COUNTIF

```
=COUNTIF(D:D,M2)
```

Counts cells matching condition.

COUNTIFS

```
=COUNTIFS($D:$D,$M12,$F:$F,N$11)
```

Multiple conditions.

SUMIF

```
=SUMIF(E:E,M21,H:H)
```

SUMIFS

```
=SUMIFS($H:$H,$G:$G,$M30,$E:$E,N$29)
```

Most commonly used in reports.

AVERAGEIF

```
=AVERAGEIF(D:D,M54,J:J)
```

AVERAGEIFS

```
=AVERAGEIFS($J:$J,$G:$G,$M63,$D:$D,N$62)
```

MAXIFS

```
=MAXIFS(J:J,C:C,M90)
```

Additional Important Statistical Functions

MINIFS

```
=MINIFS(B:B,A:A,"Finance")
```

COUNTA

Counts non-empty cells.

```
=COUNTA(A:A)
```

COUNT

Counts numeric values only.

```
=COUNT(A:A)
```

Interview Question

Difference between COUNT and COUNTA?

COUNT → Numbers only COUNTA → Non-empty cells

Chapter 8 — Date Functions

DATEDIF

```
=DATEDIF(D2,TODAY(),"Y")
```

Used for age/experience calculations.

Convert Date to Text

```
=TEXT(C11,"DDDD-MMMM-YYYY")
```

EOMONTH

```
=EOMONTH(B9,0)
```

Returns month ending date.

NETWORKDAYS.INTL

```
=NETWORKDAYS.INTL(A2,A32,1,D2:D3)
```

Calculates working days excluding holidays.

Additional Important Date Functions

TODAY

```
=TODAY()
```

NOW

```
=NOW()
```

YEAR / MONTH / DAY

```
=YEAR(A2)  
=MONTH(A2)  
=DAY(A2)
```

Chapter 9 — Pivot Table (VERY IMPORTANT)

Pivot Table is one of the MOST IMPORTANT tools for Data Analysts.

Used to summarize huge datasets quickly.

Creating Pivot Table

Steps

1. Select dataset
2. Insert → Pivot Table
3. Choose location
4. Drag fields into:
 5. Rows
 6. Columns
 7. Values
 8. Filters

Example

Analyze:

- Total sales by city
 - Average salary by department
 - Monthly profit trends
-

Pivot Table Areas

Area	Purpose
Rows	Categories
Columns	Subcategories
Values	Calculations
Filters	Global filtering

Common Pivot Calculations

- Sum
 - Count
 - Average
 - Max
 - Min
-

Pivot Table Benefits

- Fast summarization
 - No complex formulas needed
 - Interactive reporting
 - Easy filtering
 - Excellent for dashboards
-

Slicers

Interactive buttons for filtering Pivot Tables.

Real-World Usage

Used in dashboards for:

- Year filtering
 - Department filtering
 - Product filtering
-

Pivot Charts

Visual representation of Pivot Tables.

Common charts:

- Bar chart
 - Line chart
 - Pie chart
-

Excel Tables (VERY IMPORTANT)

Shortcut:

Ctrl + T

Benefits:

- Dynamic ranges
- Auto formatting
- Structured references
- Easier formulas
- Better Pivot Tables

Interview Tip

Professional analysts usually convert datasets into Tables first.

Power Query (Important Modern Skill)

Power Query is used for:

- importing data
- cleaning data
- transforming data
- combining files
- automation

Why Important?

Companies prefer automated cleaning instead of manual cleaning.

Dashboard Concepts

A dashboard is a visual summary of important KPIs.

Common Dashboard Components

- KPI cards
- Charts
- Slicers
- Pivot Tables
- Conditional Formatting

Analyst Insight

Dashboard goal: Convert raw data into business decisions.

Common Excel Interview Questions

Technical Questions

1. Difference between VLOOKUP and XLOOKUP?
2. Difference between COUNT and COUNTA?
3. Why use Pivot Tables?
4. How do you remove duplicates?
5. How do you handle missing values?
6. Difference between relative and absolute references?
7. What is conditional formatting?
8. What is data validation?

9. Difference between SUMIF and SUMIFS?
 10. Why use INDEX MATCH instead of VLOOKUP?
-

Real Data Analyst Workflow in Excel

1. Import Data
 2. Clean Data
 3. Remove duplicates
 4. Fix formatting
 5. Handle missing values
 6. Create calculations
 7. Analyze using Pivot Tables
 8. Create charts/dashboard
 9. Generate insights
 10. Share report
-

Important Excel Shortcuts

Shortcut	Purpose
Ctrl + S	Save
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + A	Select all
Ctrl + 1	Format cells
Ctrl + G	Go To
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + E	Flash Fill
Ctrl + D	Fill Down
Ctrl + R	Fill Right
Alt + =	Auto SUM
Shift + F11	New Sheet
Alt + H + V + V	Paste Values

Shortcut	Purpose
Alt + H + O + I	AutoFit Column Width
Ctrl + Arrow Keys	Move fast in dataset
Ctrl + Shift + Arrow	Select large dataset
Ctrl + T	Create Table
F4	Absolute Reference

Final Important Analyst Insights

Cleaning Data is More Important Than Fancy Charts

Most analysis problems happen because of bad data quality.

Always Check:

- duplicates
 - blanks
 - inconsistent spellings
 - extra spaces
 - wrong formats
 - missing values
-

Professional Excel Practices

- Use Tables
 - Use named ranges
 - Avoid hardcoding values
 - Use IFERROR
 - Keep raw data separate
 - Use consistent formatting
 - Document formulas when necessary
-

Final Career Advice

Excel + SQL + Power BI + Python(Pandas) is a very strong combination for Data Analyst roles.

Excel is often the FIRST tool companies expect analysts to know properly.

Your current Excel foundation is already good. The next goal should be:

- strong Pivot Tables
- dashboards
- XLOOKUP
- Power Query
- real projects
- business analysis thinking